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SIMATS ENGINEERING

ASSESSMENT TOOL 1

Experiments

COURSE NAME: Computer Networks **COURSE CODE: CSA0720**

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*
Assessment Tool Weightage: 40%
SOFTWARE: CISCO PACKET TRACER, WIRESHARK

88%

EXPERIMENTS

QUESTIONS:4 5 **Total Marks:40** 5

1. Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS). Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.
2. Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer. Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To-Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks.
3. Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the

4. Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer. Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT). Discuss how ICMP assists in network troubleshooting and connectivity verification.
5. Create a client-server network in Cisco Packet Tracer by configuring a web server and a client PC. Enable the HTTP service on the server and access the hosted webpage from the client using a web browser. Capture the communication in Wireshark and identify the TCP three-way handshake, followed by the HTTP GET request and HTTP response messages. Analyze the application-layer communication and explain how TCP provides reliable delivery for HTTP traffic through connection establishment, acknowledgments, and ordered data transfer.

Experiment 1: LAN using Switch and Wireshark

Aim

To design a simple LAN using two PCs and a switch in Cisco Packet Tracer, verify connectivity using ping, and analyze Ethernet frame fields using Wireshark.

Apparatus

- Computer
- Cisco Packet Tracer
- Wireshark
- 2 PCs
- 1 Switch
- Ethernet cables

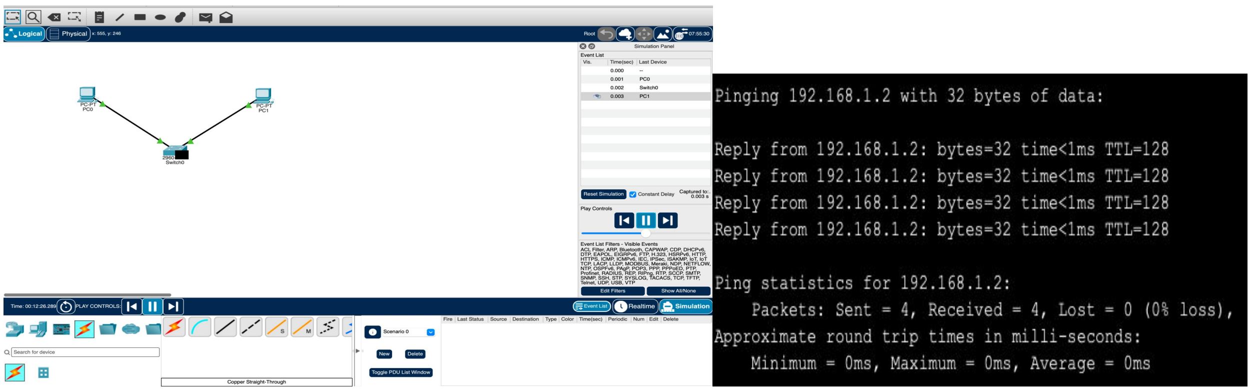
Algorithm

1. Create a LAN with two PCs and a switch.
2. Assign IP addresses to both PCs.
3. Connect the PCs to the switch.
4. Test connectivity using ping.
5. Capture the packets using Wireshark.
6. Identify the Ethernet frame fields.
7. Analyze MAC addressing and frame delivery.

Procedure

1. Open **Cisco Packet Tracer**.
2. Add **2 PCs** and **1 switch**.
3. Connect both PCs to the switch using **Copper Straight-Through** cables.
4. Configure the IP addresses:
    - PC0: 192.168.1.1
    - PC1: 192.168.1.2
    - Subnet Mask: 255.255.255.0
5. Open the command prompt on PC0.

Output:



Result

Thus, a simple LAN was successfully designed using two PCs and a switch. Connectivity was verified using the `ping` command, and Ethernet frame fields were analyzed using Wireshark. MAC addressing was used to deliver frames between devices within the local network.

Experiment 2: Two LANs Connected Through a Router

Aim

To connect two LANs using a router in Cisco Packet Tracer, configure IPv4 addresses, and verify communication using `ping`.

Apparatus

- Computer
- Cisco Packet Tracer
- 2 PCs
- 2 Switches
- 1 Router
- Ethernet cables

Algorithm

1. Create two LANs with PCs and switches.
2. Connect both switches to the router.
3. Configure IPv4 addresses and subnet masks.
4. Configure router interfaces as default gateways.
5. Test communication between the two LANs.
6. Capture ICMP packets using Wireshark.
7. Analyze the IPv4 header fields.

Procedure

1. Create **PC0 → Switch0 → Router → Switch1 → PC1**.

2. Configure PC0:

   - IP: 192.168.1.10
   - Mask: 255.255.255.0
   - Gateway: 192.168.1.1

3. Configure PC1:

   - IP: 192.168.2.10
   - Mask: 255.255.255.0
   - Gateway: 192.168.2.1

4. Configure the router interfaces:

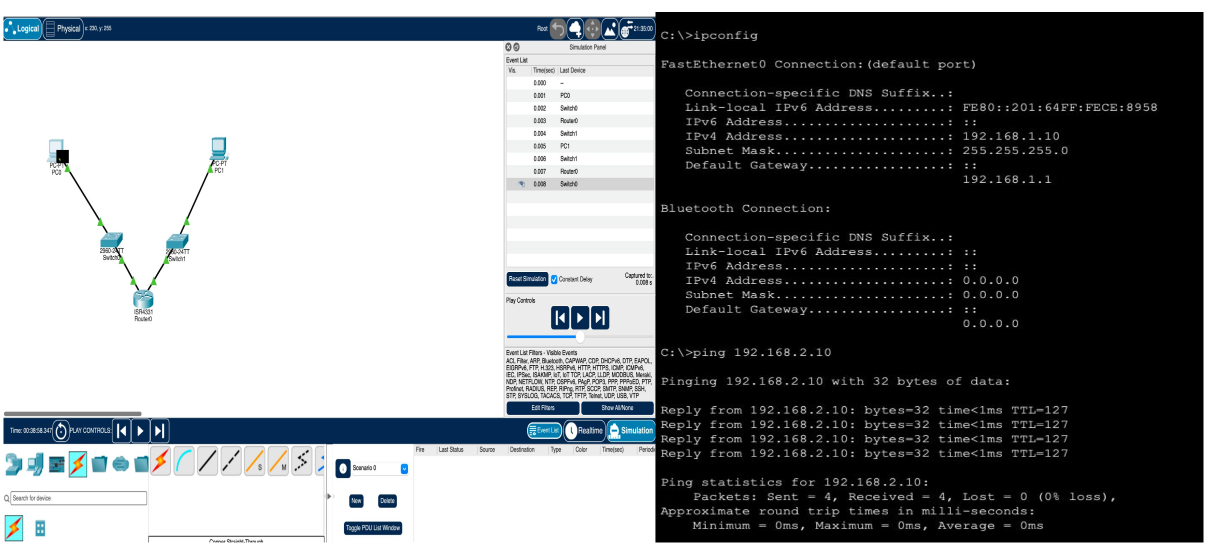
G0/0 = 192.168.1.1
G0/1 = 192.168.2.1

1. Enable the interfaces using `no shutdown`.
2. From PC0, execute:

```
ping 192.168.2.10
```

1. Capture the ICMP packets in Wireshark.
2. Examine Source IP, Destination IP, TTL, Protocol and Header Length.

Output:



Result

Thus, two LANs were successfully connected through a router, and end-to-end communication was verified using `ping`. IPv4 header fields were successfully analyzed.

Experiment 3: ARP Request and Reply Analysis

Aim

To analyze ARP Request and ARP Reply packets and understand how IP addresses are resolved into MAC addresses.

Apparatus

- Computer
- Cisco Packet Tracer

- Wireshark
- 2 PCs
- 1 Switch
- Ethernet cables

Algorithm

1. Create a LAN using two PCs and a switch.
2. Configure IP addresses.
3. Clear the ARP cache.
4. Ping the destination PC.
5. Capture the packets using Wireshark.
6. Identify ARP Request and ARP Reply.
7. Analyze sender and target MAC/IP addresses.

Procedure

1. Connect two PCs to a switch.
2. Configure:

PC0: 192.168.1.10
PC1: 192.168.1.20
Subnet Mask: 255.255.255.0

1. Clear the ARP cache on PC0.
2. Ping PC1:

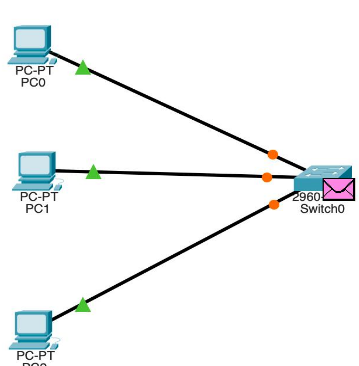
ping 192.168.1.20

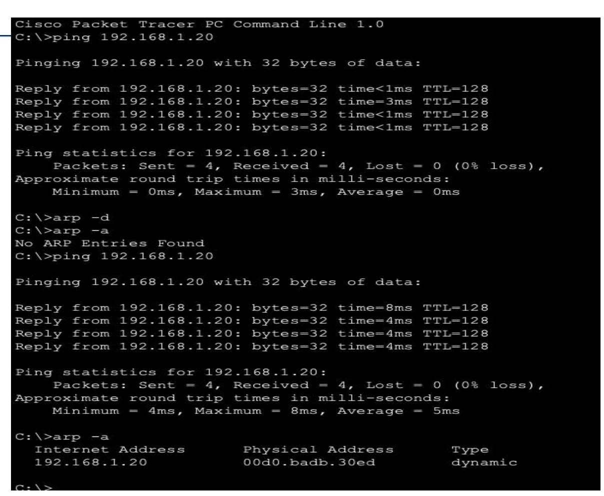
1. Start Wireshark capture.
2. Apply the filter:

arp

1. Observe the **ARP Request**.
2. Observe the **ARP Reply**.
3. Note the sender and target MAC/IP addresses.

Output:





Result

Thus, ARP Request and Reply packets were successfully captured and analyzed. ARP resolved the destination IP address into its corresponding MAC address.

Experiment 4: ICMP Echo Request and Echo Reply

Aim

To analyze ICMP Echo Request and Echo Reply messages and calculate the Round Trip Time (RTT).

Apparatus

- Computer
- Cisco Packet Tracer
- Wireshark
- 2 PCs
- 1 Router
- Ethernet cables

Algorithm

1. Connect two PCs through a router.
2. Configure IP addresses and gateways.
3. Ping the destination PC.
4. Capture ICMP traffic using Wireshark.
5. Identify Echo Request and Echo Reply.
6. Analyze Type, Code, Identifier and Sequence Number.
7. Calculate RTT.

Procedure

1. Configure PC0:

```
IP: 192.168.1.10
Gateway: 192.168.1.1
```

1. Configure PC1:

```
IP: 192.168.2.10
Gateway: 192.168.2.1
```

1. Configure the router interfaces.
2. From PC0, execute:

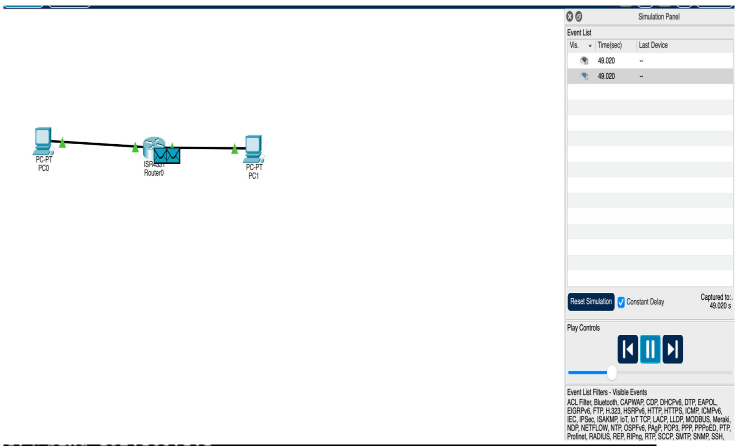
```
ping 192.168.2.10
```

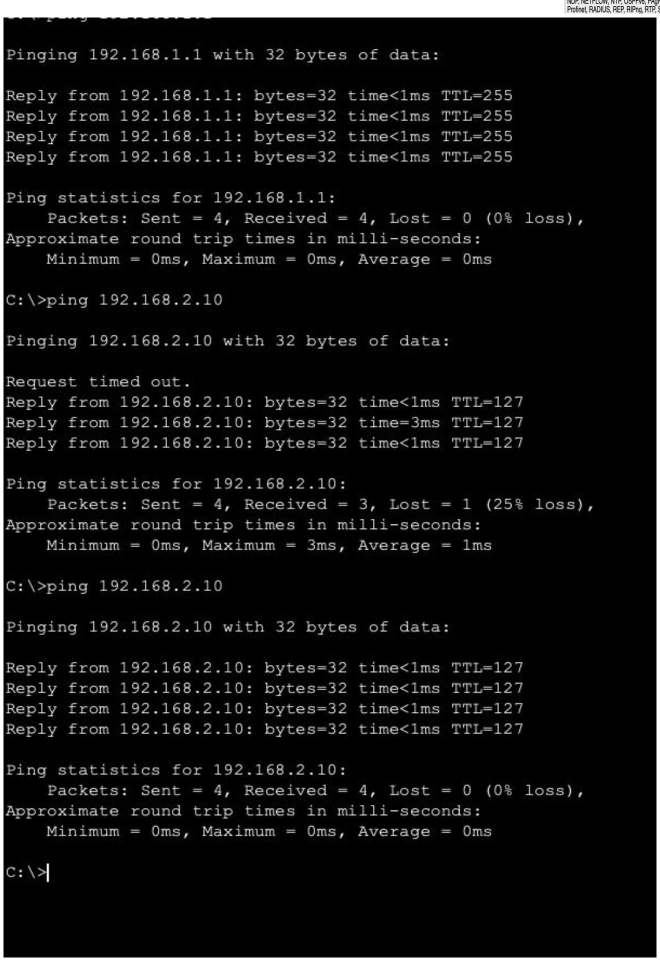
1. Capture packets in Wireshark.
2. Apply:

```
icmp
```

1. Identify **Echo Request** and **Echo Reply**.
2. Observe Type, Code, Identifier, Sequence Number and RTT.

Output:





Result

Thus, ICMP Echo Request and Echo Reply messages were successfully captured and analyzed. The RTT was obtained, demonstrating the use of ICMP for connectivity testing and network troubleshooting.

Experiment 5: Client-Server HTTP Communication

Aim

To create a client-server network, access a web server from a client PC, and analyze the TCP three-way handshake and HTTP communication using Wireshark.

Apparatus

- Computer
- Cisco Packet Tracer
- Wireshark
- 1 PC
- 1 Switch

- 1 Server
- Ethernet cables

Algorithm

1. Create a network with a PC, switch and server.
2. Configure IP addresses.
3. Enable HTTP service on the server.
4. Access the webpage from the client.
5. Capture packets using Wireshark.
6. Identify TCP three-way handshake.
7. Identify HTTP GET and HTTP response.

Procedure

1. Connect **PC → Switch → Server**.
2. Configure PC:

IP: 192.168.1.10
Subnet Mask: 255.255.255.0

1. Configure Server:

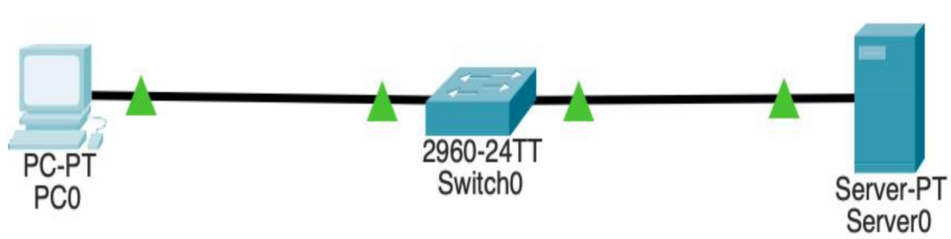
IP: 192.168.1.20
Subnet Mask: 255.255.255.0

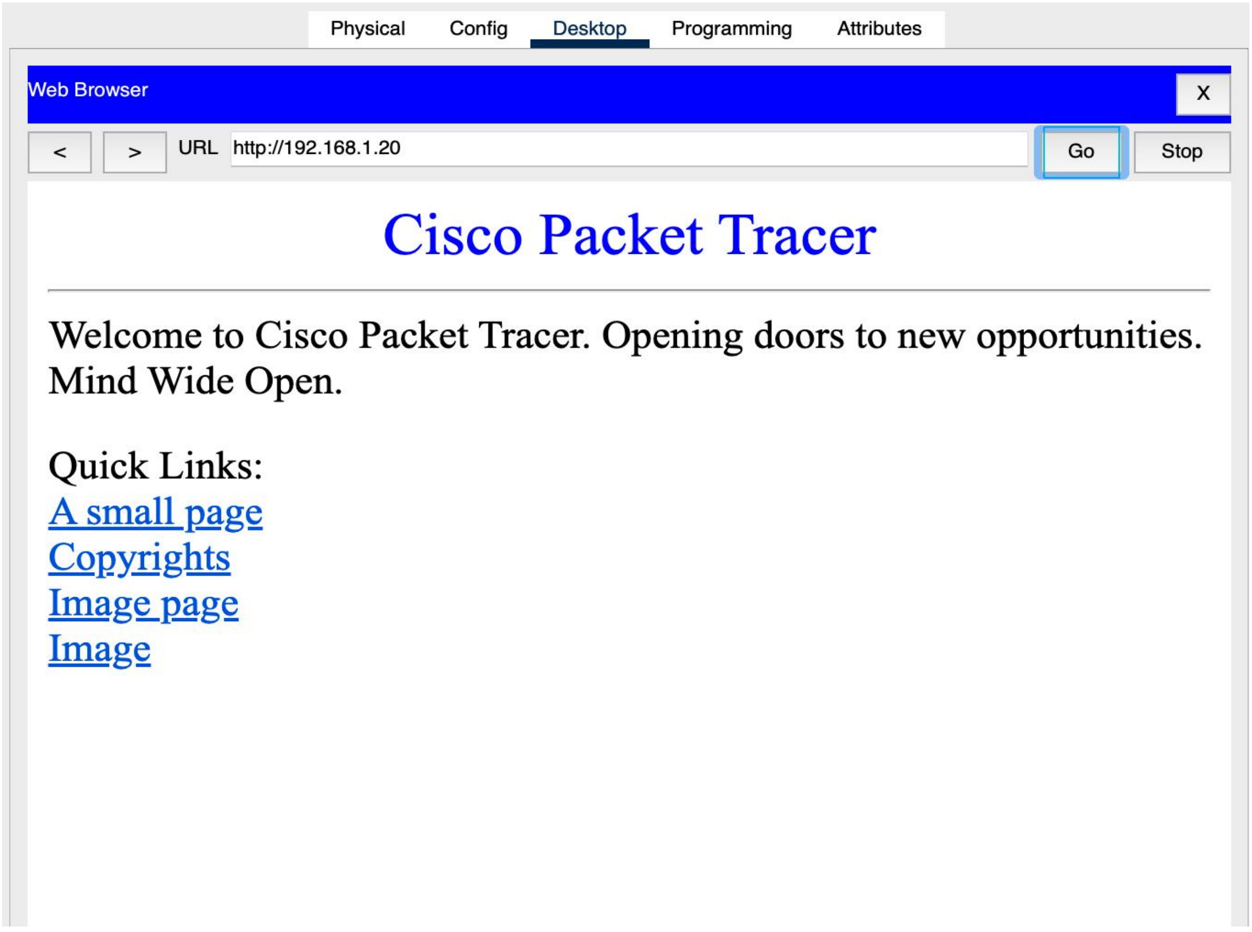
1. Open **Server → Services → HTTP**.
2. Turn **HTTP ON**.
3. Open the PC's **Web Browser**.
4. Enter:

http://192.168.1.20

1. Verify that the webpage loads.
2. Capture the traffic using Wireshark.
3. Analyze **SYN → SYN-ACK → ACK**, followed by **HTTP GET → HTTP Response**.

Output:





Result

Thus, the client-server HTTP network was successfully configured. The TCP three-way handshake, HTTP GET request, and HTTP response were captured and analyzed, demonstrating reliable TCP communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results

		interpretation and validation of results	reasonable interpretation	limited insights	analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

	Understanding of Concepts **(25%)**	Experimental Design **(30%)**	Use of Tools **(30%)**	Analysis of Results **(15%)**	Documentation **(10%)**
Q1	2	3	2	1.5	1
Q2	2	3	1.5	1.5	1
Q3	2.5	2.5	2	1	0.5
Q4	2.5	2	1.5	1.5	1
Q5	2	3	2	1	0.5

AA